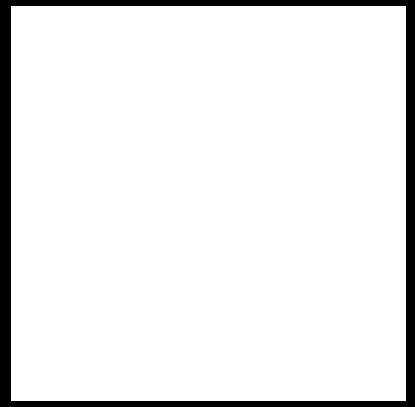
ual: creative computing institute

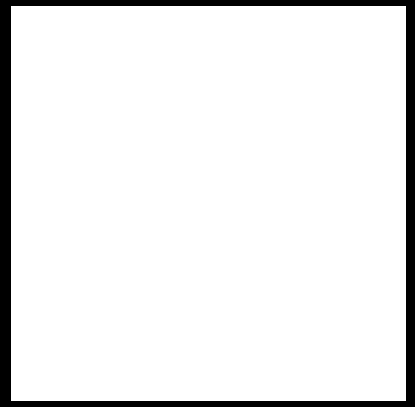
Generative Text in Performance - Week 2

Performing Data 25-26

Louis McCallum



Generative Text and Human Collaborations



Opulent AI

Kate Compton



Poor little Artificial Intelligence

She scurries around behind the scenes of games, behind Google Maps and page rankings.
"Find me the fastest path, AI!"
"Solve this problem for me, AI!"
"Route this NPC to the end of the level without running into walls, AI!"
"But do so UNOBTRUSIVELY. Do so UNSEEN. Never, never, make yourself a spectacle. They do not wish to see you or talk to you. You are just here to do work."

"....Oh, and can you do it faster and with less resources?"

HOW DREADFUL
lets make her FABULOUS!



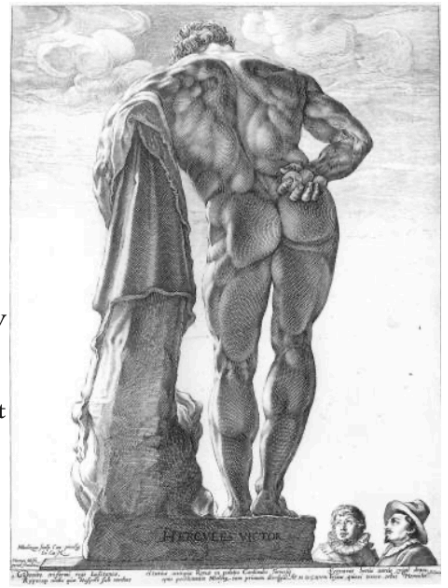
I want to coin the term "Opulent AI" for AI that takes up **all the resources** it wants, takes **all the attention**, and makes the experience **all about itself**.

FOR NO PRACTICAL PURPOSE WHATSOEVER

Visualization: Lay it all bare

What can the user learn about how the AI works? Lay its glorious operations bare before them so that they may marvel at its construction!

Approaches:
Show visualizations of what your AI's state currently is, the more shiny and graphical the better!
Let your AI say what its thinking. Why did it make those decisions? Let it expound upon its sagacity and wisdom.
Add hocks that let you track and output the logic that got it to those decisions
Open it up! Leave the backdoor to its brain open, leave the rules and datafiles in plaintext. Allow the suers to admire them, edit them, and reshare them.



A few Opulent AI systems:

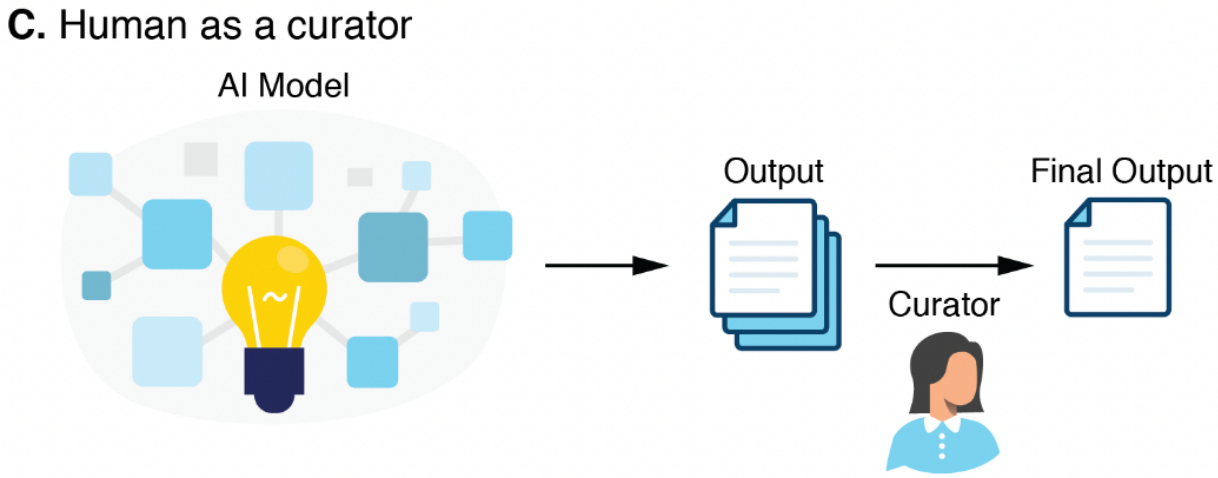
Black and White, the Sims, Nicki Case's Simulating the World in Emoji (http://ncase.me/simulating/), In-groups & Outgroups http://ncase.me/group-prototype/, Vi Hart and Nicki Case's http://ncase.me/polygons/, Nicki Cases' Neurotic Neurons http://ncase.me/neurons/, Context Free art http://www.contextfreeart.org/, Tracery (tracery.io) and the many Tracery and

http://www.galaxykate.com/pdfs/galaxykate-zine-opulentai.pdf

Human in the Loop

Human as Curator

- Often, we have seen the “human as curator” approach to creative projects with text generation

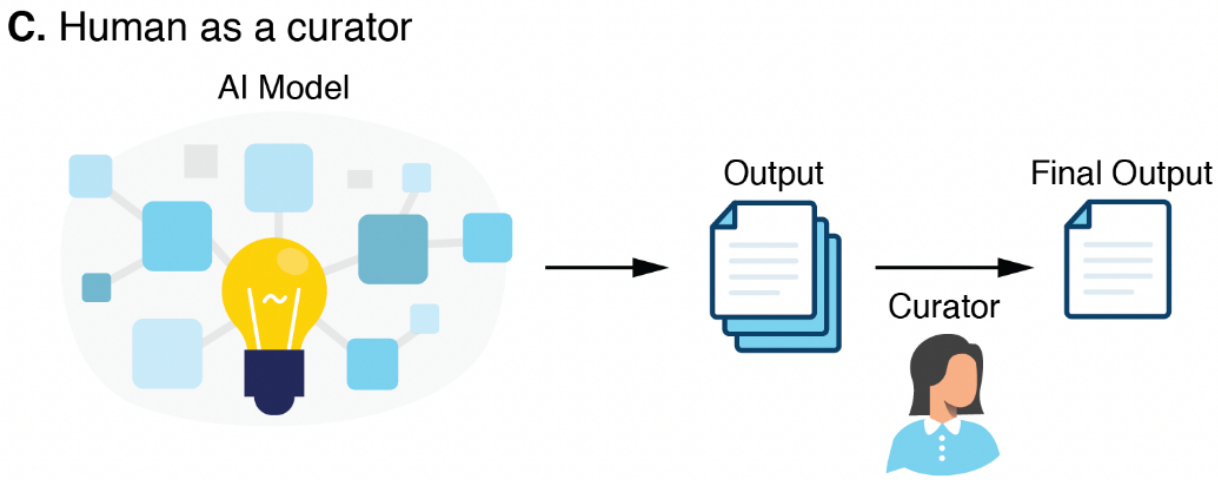


Chung, N.C., 2021. Human in the Loop for Machine Creativity.

Human in the Loop

Human as Curator

- This tends to see the generation of the final text as the end point

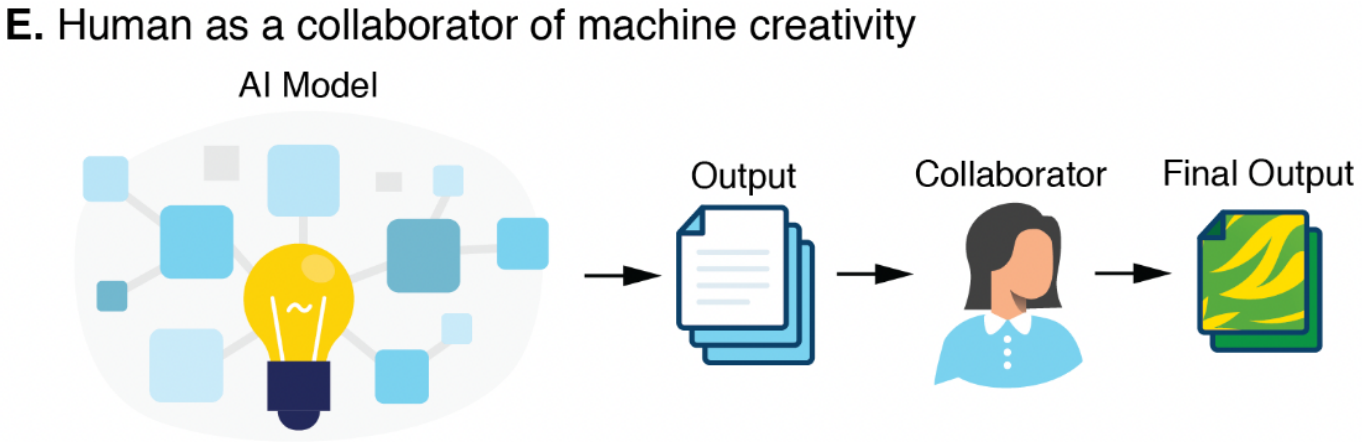


Chung, N.C., 2021. Human in the Loop for Machine Creativity.

Human in the Loop

Human as Collaborator

- However, we will now see how artists and musicians can take this generated text and use it to further prompt creative work, or present the outcome in creative ways



Chung, N.C., 2021. Human in the Loop for Machine Creativity.

Human in the Loop

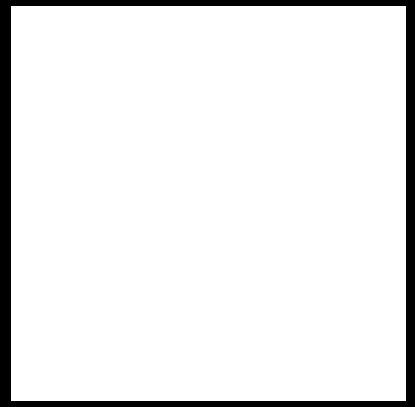
Human as Collaborator

Viq & I (2022) - Eric Drass

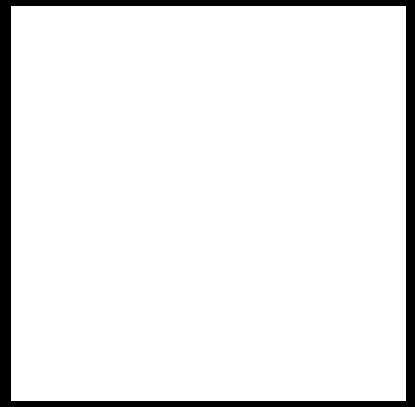


Text to Everything

- Generated text now routinely feeds into other generative systems: text-to-image, text-to-music (MusicLM, Udio), text-to-video (Sora), text-to-3D
- This changes the role of generated text from final output to intermediate instruction as language becomes a control interface for all media



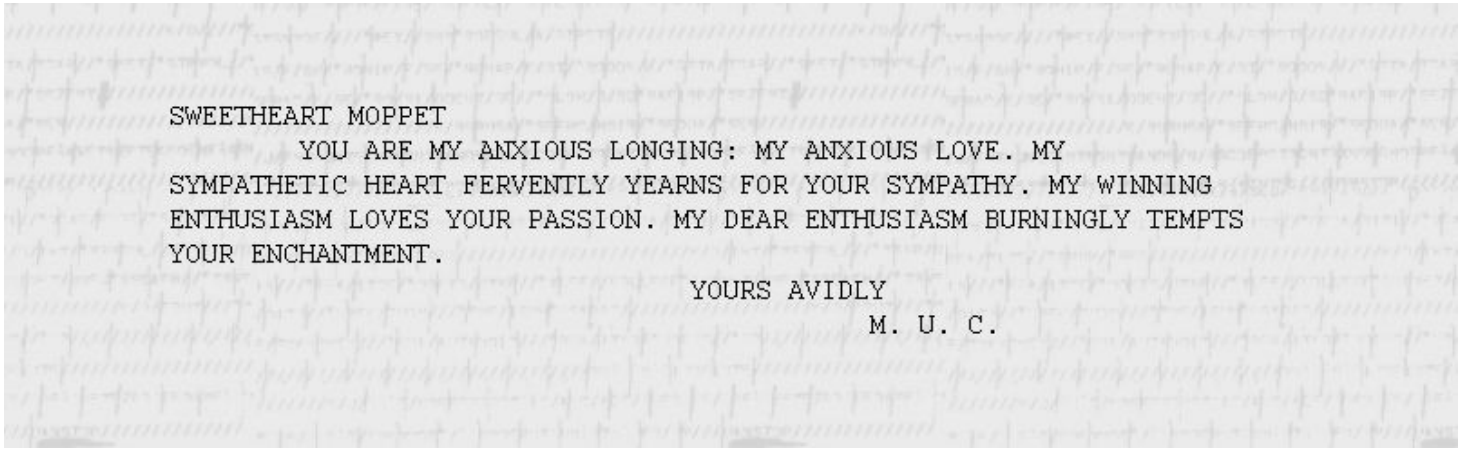
Template Based Text Art



Strachey Love Letter Generator

The first computer text artist?

- “…beginning in August of 1953, short notes began appearing on the notice board of the Manchester University Computer Department. They appeared to be letters of love and adoration addressed to an unnamed, genderless other, signed only with the initials M.U.C.”



SWEETHEART MOPPET
YOU ARE MY ANXIOUS LONGING: MY ANXIOUS LOVE. MY
SYMPATHETIC HEART FERVENTLY YEARNS FOR YOUR SYMPATHY. MY WINNING
ENTHUSIASM LOVES YOUR PASSION. MY DEAR ENTHUSIASM BURNINGLY TEMPTS
YOUR ENCHANTMENT.
YOURS AVIDLY
M. U. C.

https://rhizome.org/editorial/2013/apr/9/queer-history-computing-part-three/

Strachey Love Letter Generator

The first computer text artist?

- Generated on the Manchester University Computer (M.U.C) based on the following template
- You are my — Adjective (adj) — Substantive (noun),”
- Repeat
    - “My — [Adjective] — Substantive — [Adverb (adv)] — Verb (verb) — Your — [Adjective] — Substantive.]
- Square brackets indicate optional inclusion

http://www.alpha60.de/art/love_letters/DavidLink_DasHerzDerMaschine_2012.pdf

Strachey Love Letter Generator

The first computer text artist?

- A simple rule for quite moving letters
- 318 billion combinations
- Is this a bleak representation that love can be computerised? Or does it confirm love is a projection from the reader?
- Jacob Garboury's A Queer History of Computing has some really interesting thoughts

https://rhizome.org/editorial/2013/apr/9/queer-history-computing-part-three/

Strachey Love Letter Generator

An absence of “love”?

- Some people have noted (and drawn conclusions) from the fact none of the existing examples included the word “love”
- However, further research has determined the original dictionary of terms did include “several variations on the word love, there simply were no examples of such letters in wide circulation”

https://rhizome.org/editorial/2013/apr/9/queer-history-computing-part-three/

Strachey Love Letter Generator

Try it out online

CHRISTOPHER STRACHEY "LOVELETTERS" (1952)

DEAREST DUCK,
YOU ARE MY PRECIOUS DESIRE, MY FONDEST LONGING. MY DESIRE WOOS YOUR CRAVING AMBITION. MY LOVEABLE PASSION LIKES YOUR BEING. YOU ARE MY LOVESICK DESIRE.
YOURS IMPATIENTLY,
M.U.C.

GENERATE ANOTHER LOVELETTER

SALUTATIONS1	SALUTATIONS2	ADJECTIVES	NOUNS	ADVERBS	VERBS
BELOVED	CHICKPEA	AFFECTIONATE	ADORATION	AFFECTIONATELY	ADORES
DARLING	DEAR	AMOROUS	AFFECTION	ARDENTLY	ATTRACTS
DEAR	DUCK	ANXIOUS	AMBITION	ANXIOUSLY	CLINGS TO
DEAREST	JEWEL	AVID	APPETITE	BEAUTIFULLY	HOLDS DEAR
FANCIFUL	LOVE	BEAUTIFUL	ARDOUR	BURNINGLY	HOPES FOR
HONEY	MOPPET	BREATHLESS	BEING	COVETOUSLY	HUNGERS FOR
	SWEETHEART	BURNING	BURNING	CURIOUSLY	LIKES
		COVETOUS	CHARM	EAGERLY	LONGS FOR
		CRAVING	CRAVING	FERVENTLY	LOVES
		CURIOUS	DESIRE	FONDLY	LUSTS AFTER
		EAGER	DEVOTION	IMPATIENTLY	PANTS FOR
		FERVENT	EAGERNESS	KEENLY	PINES FOR
		FONDEST	ENCHANTMENT	LOVINGLY	SIGHS FOR
		LOVEABLE	ENTHUSIASM	PASSIONATELY	TEMPTS
		LOVESICK	FANCY	SEDUCTIVELY	THIRSTS FOR
		LOVING	FELLOW FEELING	TENDERLY	TREASURES
		PASSIONATE	FERVOUR	WISTFULLY	YEARNS FOR
		PRECIOUS	FONDNESS		WOOS
		SEDUCTIVE	HEART		

https://www.gingerbeardman.com/loveletter/

Tracery

- Kate Compton (who you saw in the Opulent AI zine) also created Tracery, a widely-used tool for generative text using context-free grammars
- Sits between the rigid template approach of Strachey and the "black box" of neural networks
- Hugely popular in game design, Twitter bots, and creative coding education
- http://tracery.io/

TV Comments Bot

David Lublin

- Replaces original subtitles with new ones in a way that is really nicely surreal
- Image recognition looks for what's in the frame, and a "madlibs" approach is used to swap in the nouns of the new objects

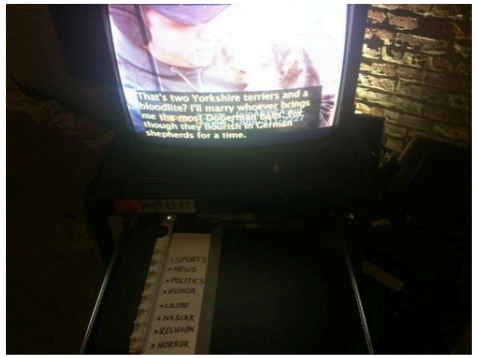




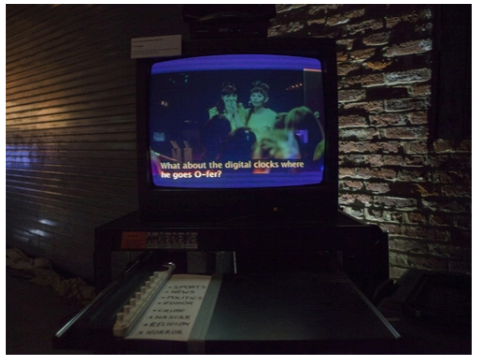
TV Comments Bot

David Lublin

- Was presented as an TV channel helper as an installation
- Also exists as a twitter bot (@TVCommentsBot)







Allison Parrish Computational Poetry

- Allison Parrish is a poet and programmer whose work uses word vectors and semantic space rather than just training on existing text
- Instead of asking "what word comes next?" she asks "what words are nearby in meaning-space?"
- Parrish deliberately designs systems that produce surprising language
- Works include "Articulations" (2018), generated from Project Gutenberg texts using phonetic and semantic similarity
- https://www.decontextualize.com/

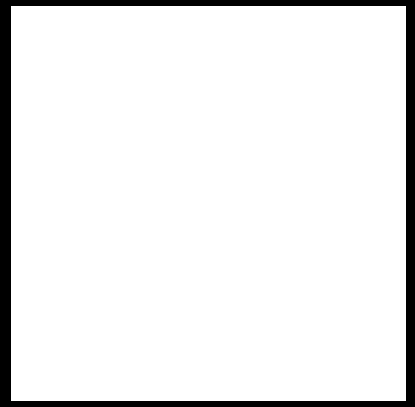


1 the Road Ross Goodwin (2018)

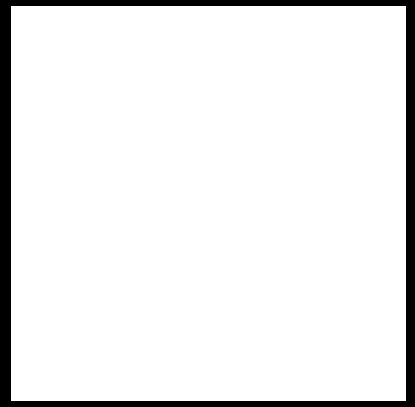
- A neural network was installed in a car with a camera, GPS, clock, and microphone as inputs
- As the car drove from New York to New Orleans, the system generated a novel in real-time, printed on a receipt printer inside the vehicle
- Deliberately echoing Jack Kerouac's "On the Road" but its a machine responding to its environment
- Raises questions about authorship, site-specificity, and what it means for text generation to be embodied and performed in real-time

1 the Road Ross Goodwin (2018)





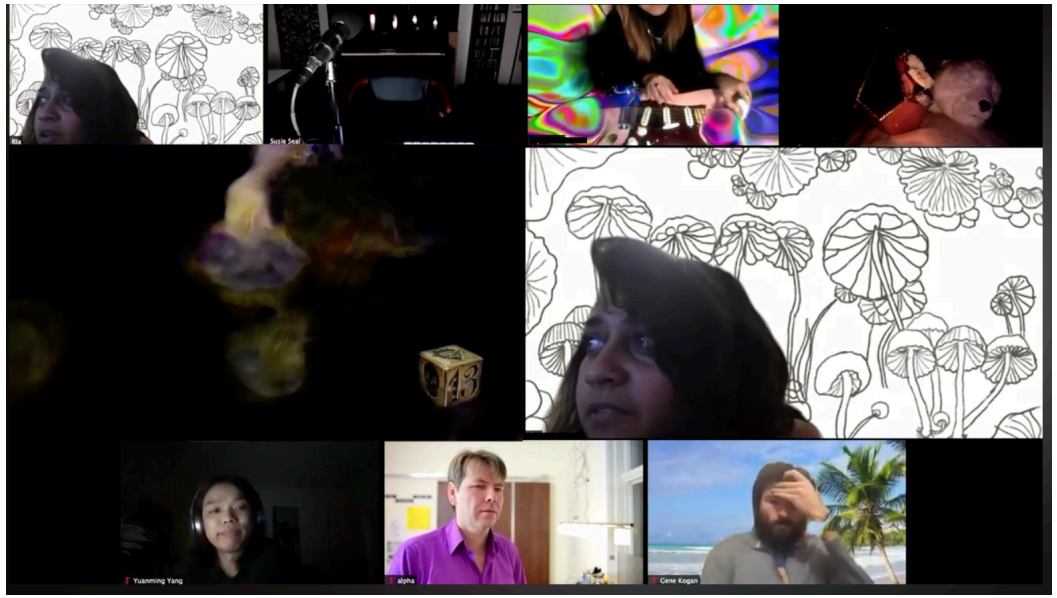
Words and Music



The Seals

Yuditskaya, S., Schedel, M., Sun, S., Rajan, R., & Green, S. (2021). The Seals International Debut. NIME 2021

- We are a political, feminist, noise and AI inspired electronic sorta-surf rock band composed of Meg, Sofy, Sophia, Ria, Susia, and the SEAL (Synthetic Erudition Assist Lattice) AIs (the collection of AIs that assist us in creating usable content with which to mold and shape our music and visuals).
-

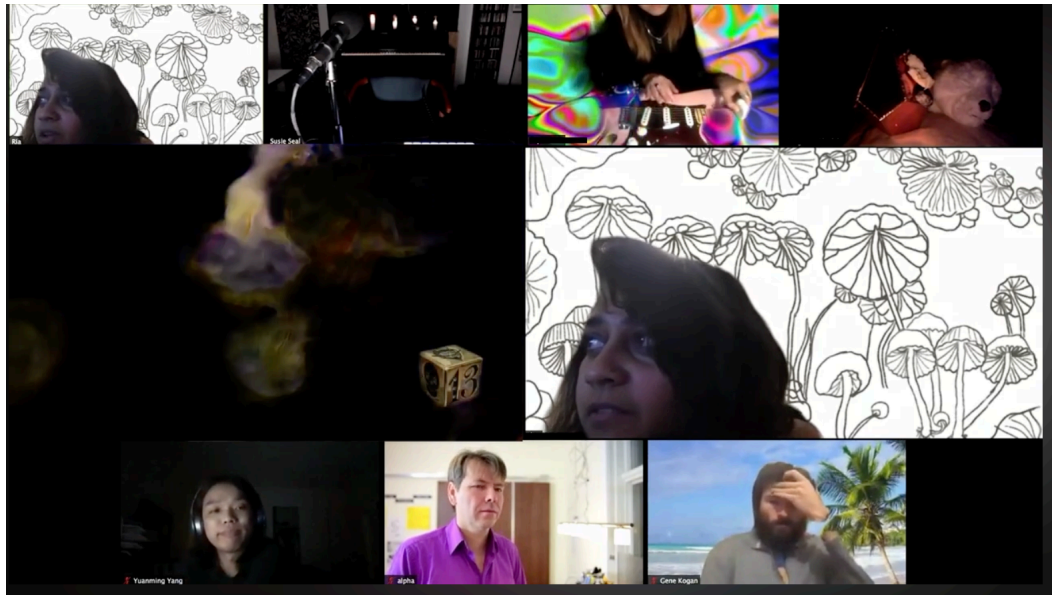


https://nime.pubpub.org/pub/m0a5jsok/release/1?readingCollection=bd12ca41

The Seals

Yuditskaya, S., Schedel, M., Sun, S., Rajan, R., & Green, S. (2021). The Seals International Debut. NIME 2021

- We begin by invoking one another through zoom and play skull theremins while reading GPT2 & GPT3 (Machine Learning) generated dialogue



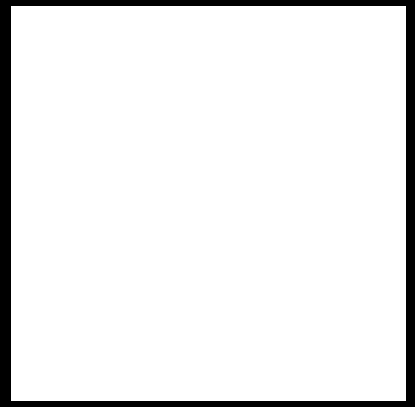
https://nime.pubpub.org/pub/m0a5jsok/release/1?readingCollection=bd12ca41

AI Song Contest

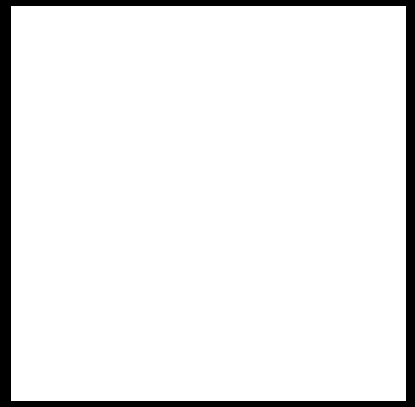
2020-Present

- The AI Song Contest was initiated and organised for the first time in 2020 by Dutch public broadcaster VPRO to run alongside the Eurovision Song Contest
- It has since continued inviting bands to submit songs involving AI co-creation
- They are relevant to NLP because lots of contestant use AI to generate, or inspire, lyrics which are then often performed by humans in submissions
- The contest has a nice looseness in how much AI much be used (e.g. they are not dogmatic about removing humans completely!)

https://www.aisongcontest.com/



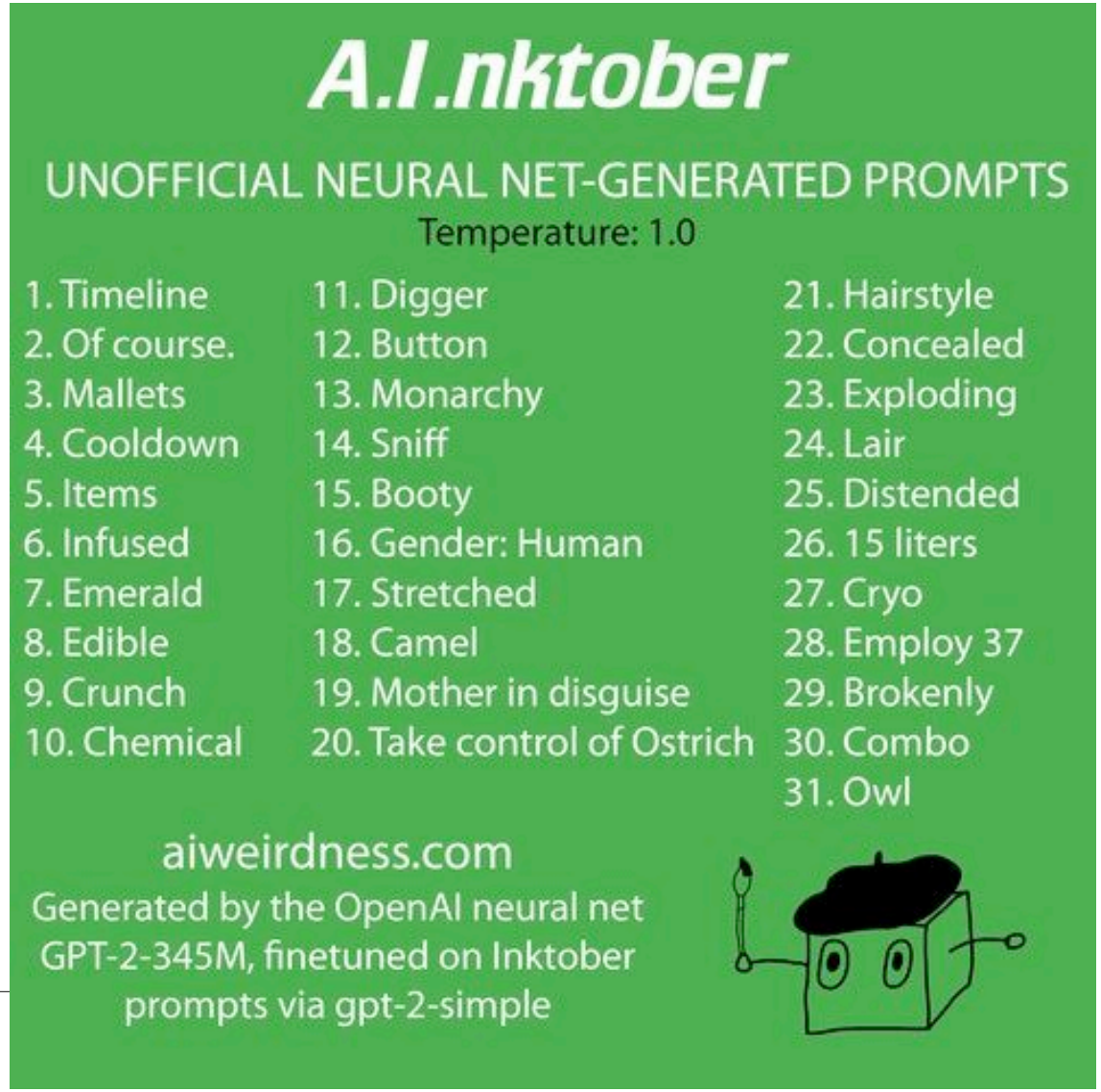
Performing AI



A.I.nktober

Janelle Shane

- Inktober normally runs setting daily drawing prompts throughout October
- Janelle Shane took this data and fine-tuned a model to generate new prompts
- Insanely, given the quality of text to image in 2026, in 2019 she says “One of these days I’m hoping for a neural net that can participate in Inktober itself” !



https://www.aiweirdness.com/tag/inktober/

AI in films

As Characters

- Often used as characters
- See 1927 film *Metropolis* often cited as first of the genre
- But we will look at AI as generators!

1927	*Metropolis*	Germany	Maria's robot double
1934	*Der Herr der Welt* (i.e. *Master of the World*)	Germany	Überroboter / Kamp machine), working-r
1951	*The Day the Earth Stood Still*	USA	Gort
1957	*The Invisible Boy*	USA	Robby the Robot
1962	*The Creation of the Humanoids*	USA	✓
1965	*Alphaville*	France	Alpha 60
1968	*2001: A Space Odyssey*	UK / USA	HAL 9000
1970	*Colossus: The Forbin Project*	USA	Colossus, Guardian
1972	*Silent Running*	USA	Huey, Dewey and L
	Westworld	USA	
	Demon Seed	USA	Proteus IV
	Star Wars	USA	R2-D2, C-3PO
	Star Trek: The Motion Picture	USA	V'Ger



https://en.wikipedia.org/wiki/List_of_artificial_intelligence_films

Sunspring (2016)

ArsTechnica

- Trained a CharRNN on sci fi film scripts, generated a new one
- Got actors to interpret and filmed the short film (which we will watch and review later!)
- There have been many AI text generation projects (including film, poetry etc..), but taking the time to film it, and in music, really adds to it through some layers of human interpretation

https://arstechnica.com/gaming/2021/05/an-ai-wrote-this-movie-and-its-strangely-moving/

Sunspring

ArsTechnica

- “There's an interesting recurring pattern in Sunspring where characters say, 'No I don’t know what that is. I’m not sure,'" said Goodwin. "They're questioning the environment, questioning what’s in front of them. There's a pattern in sci-fi movies of characters trying to understand the environment.”



https://arstechnica.com/gaming/2021/05/an-ai-wrote-this-movie-and-its-strangely-moving/

Sunspring

ArsTechnica

- They replaced character names with one letter symbols (extending consistent character arcs over the whole script was too hard otherwise, due to the size of CharRNNs window)

https://arstechnica.com/gaming/2021/05/an-ai-wrote-this-movie-and-its-strangely-moving/

Algonory

Eric Drass (Shardcore)

- Generating new stories based on children's authors
- Fine tuning of GPT2
- Filming the result in the style of Jackanory (ran on BBC television from 1965 until 1996)
- Again, adds an extra element of human interpretation and performance



https://www.shardcore.org/shardpress2019/2019/06/27/algonory/

Date Night (2021)

Calamity AI

- Similar idea to Sunspring, but made later and using GPT-3
- Generated a script using a short prompt (instead of using a training corpus of film scripts)
- Then filmed it

> INT. BENNY'S HOUSE – NIGHT
> BENNY and JULIA (BOTH 20'S) sit at two ends of a long table, drinking wine.
> BENNY: I wanna play a li'l game. It's a li'l crazy, it's a li'l fun, but it'll blow your fucking mind.

https://builtin.com/media-gaming/ai-movie-script

Date Night (2021)

Calamity AI

- Generated 3 versions of the script but threw away the first two, one because it had strong themes of misogynistic violence
- This seems to track with a lot of people's experience of GPT-3



https://builtin.com/media-gaming/ai-movie-script

Date Night (2021)

Calamity AI

- You would have expected GPT-3 to have better long term structure, which it does to a certain extent
- They actually end up talking about the fact the are in a film

JULIA
Benny. Are you sure we’re real?

BENNY
I don't know. Are you sure you're real?

JULIA
No one tells me. They just write what I do . I don't know what I’m… I just know there’s a script somewhere and people come up with

BENNY
I can't believe we're making a movie. Now, remind me, what's it about?

JULIA
We're a couple. It's a relationship movie.

BENNY
What’s the movie called?

JULIA
“Date Night.” He hypnotizes her. It gets dark. He turns her face into his.

BENNY
“Date Night.”

JULIA
“Date Night.” “Date Night.”

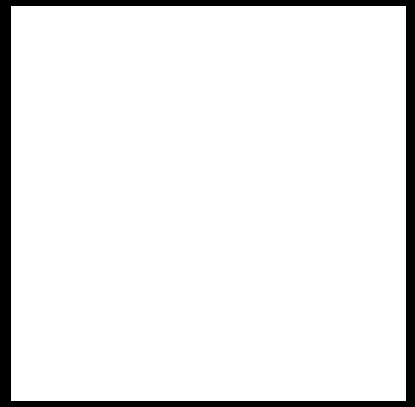
https://drive.google.com/file/d/1LRqlWNN3II0e3H9v2Iy2mJAmXv9MIhoZ/view

A Piece of Work - Annie Dorsen (2013)

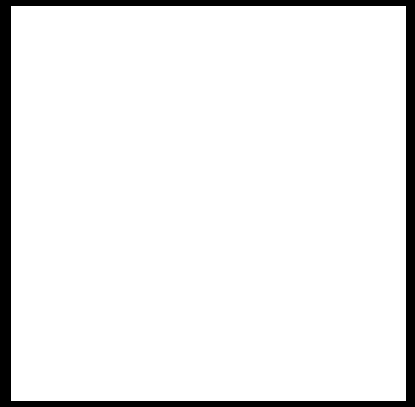
- An algorithmic performance of Hamlet where software recombines and mutates Shakespeare's text live on stage
- No two performances are the same as the algorithm makes different choices each night
- Actors must respond to and interpret text they have not seen before, similar to the Sunspring actors interpreting a CharRNN script
- Extends the "human as collaborator" model into live theatre, where the liveness of the performance adds another layer of unpredictability
- http://anniedorsen.com/

A Piece of Work - Annie Dorsen (2013)





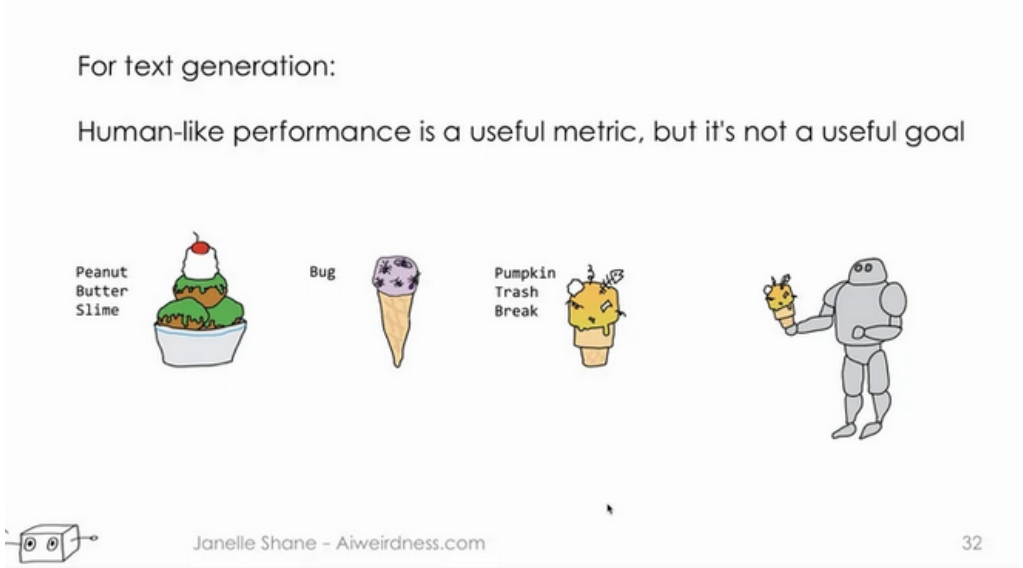
Into the GPT



Farewell to Fart

Janelle Shane

- Janelle Shane has an excellent talk on the difference in working with CharRNN (Sunspring) and GPT-3 (Date Night)
- She says the big language models are good at being cliche and realistic, but this is not necessarily good for art or humour
- She has to work harder to be weird
- Also calls working with GPT-3 “traumatic” (because of the explicit stuff it comes out with)

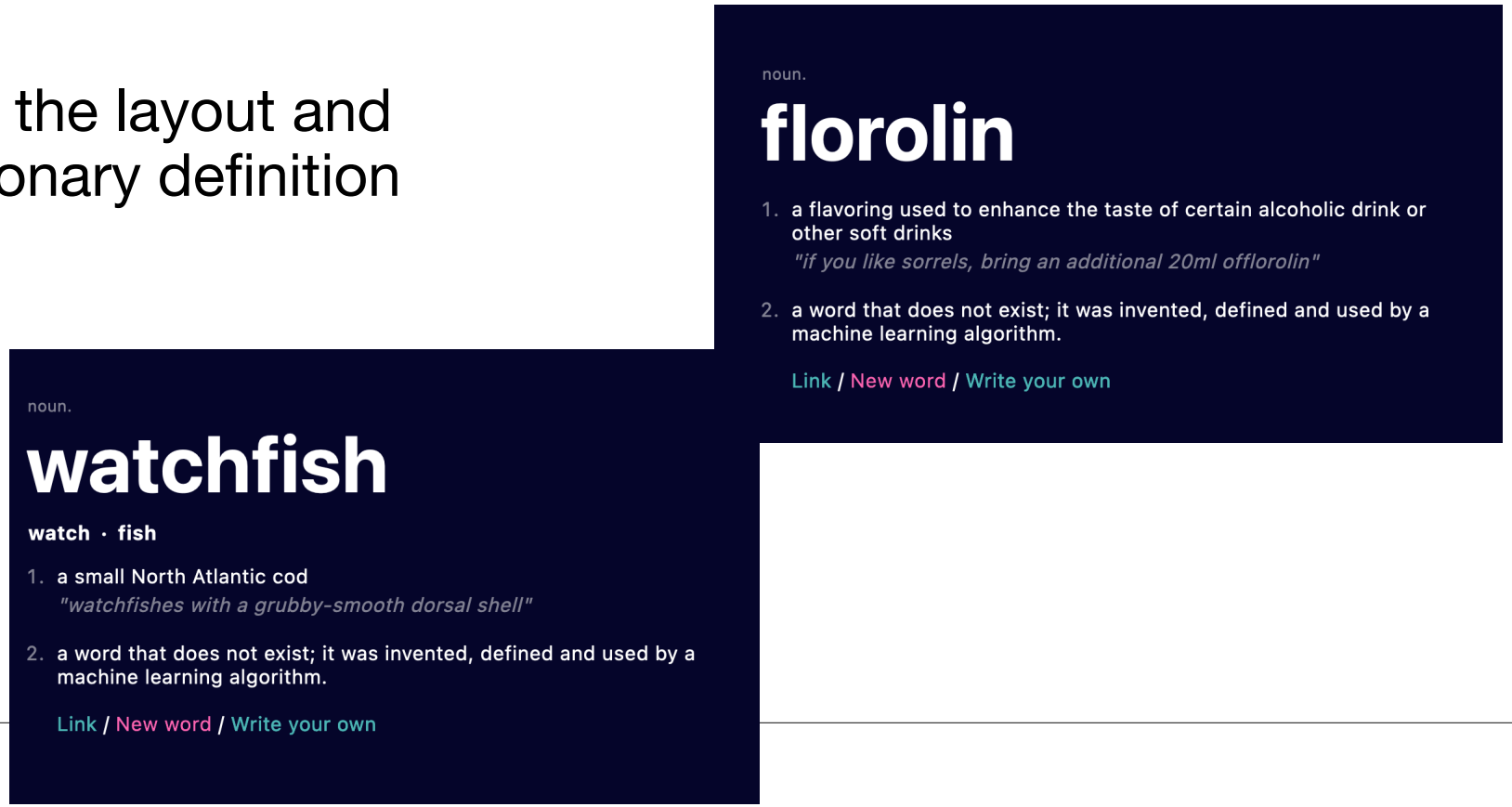


https://crossminds.ai/video/farewell-to-fart-working-with-generated-text-in-the-age-of-huge-neural-nets-606fdb85f43a7f2f827bf4eb/

Neologisms

Thomas Dimson (@turtlesoupy)

-
- Taking care to ape the layout and structure of a dictionary definition



Pharmako-AI by K Allado-McDowell (2020)

- A book co-written with GPT-3 where the human author and the AI alternate writing passages
- Neither voice is clearly dominant
- One of the earliest and most thoughtful examples of the "human as collaborator" model applied to long-form writing
- Published by Ignota Books, it was one of the first commercially published human-AI co-authored books
- Interesting to compare with the curation model as here the human does not simply select from AI outputs but actively responds to them



AI Dungeon - Latitude (2019)

- One of the first widely-used interactive fiction experiences powered by GPT-2, later GPT-3
- Players type actions in natural language and the AI generates the story in response — the human is simultaneously performer, audience, and co-author
- Blurs the human-in-the-loop categories: the player is both curator (choosing which outputs to accept) and collaborator (actively shaping the narrative)
- Also encountered significant content moderation challenges, echoing the Date Night experience of GPT-3 producing disturbing content



- https://aidungeon.com/

Nothing, Forever by Mismatch Media (2023)

- A never-ending AI-generated sitcom that streamed continuously on Twitch, styled after Seinfeld
- Characters, dialogue, and plot were all generated in real-time by language models, with 3D animated characters delivering the lines
- Gained a large audience, but was temporarily banned from Twitch when the model generated transphobic content during a live stream
- Raises the same issues as Date Night's discarded scripts, but amplified — with no human curator in the loop, problematic content goes directly to an audience
- A case study in what happens when you remove the "human as curator" stage entirely



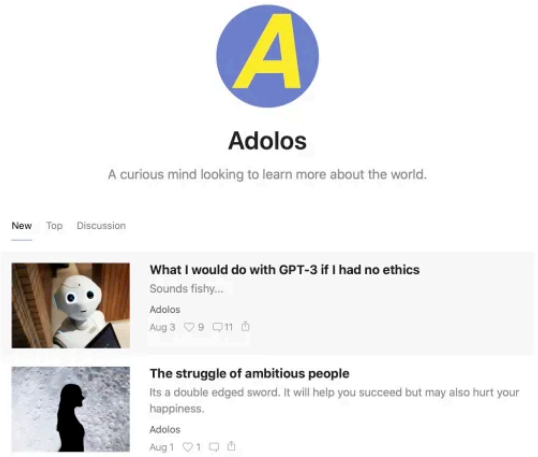
https://www.twitch.tv/watchmeforever

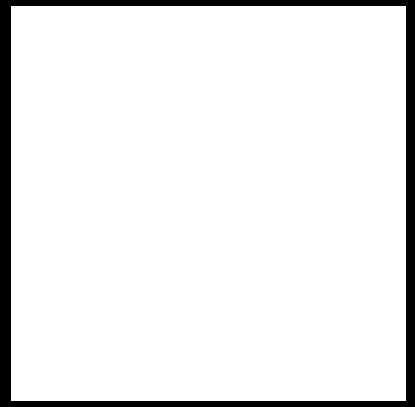
LAUREN Lauren Lee McCarthy

- Lauren Lee McCarthy offered to be a human-powered smart home assistant — replacing Alexa/Siri with a real person watching and responding through connected devices
- Inverts the usual human-AI performance relationship: instead of AI pretending to be human, a human pretends to be AI
- Participants knew a human was controlling their home but still found themselves talking to her as though she were a device
- Relevant to thinking about how generated text is received — do we respond differently when we know the source?
- https://lauren-mccarthy.com/LAUREN

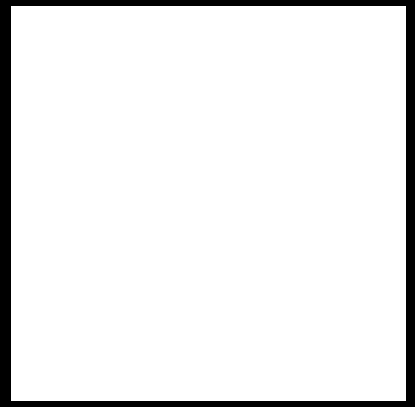
Liam Porr's GPT-3 Blog Authorship and Deception (2020)

- A UC Berkeley student used GPT-3 to write blog posts and published them under a fake name on a blogging platform
- One post reached the top of Hacker News — very few readers noticed it was AI-generated
- Raises ethical questions about transparency and authorship in creative AI text — when does collaboration become deception?
- Interesting contrast with Sunspring and Algonory, where the AI authorship is foregrounded and celebrated as part of the work
- The creative framing makes all the difference -> the same generated text can be art, satire, or fraud depending on context





Text as Instruction



Fluxus Event Scores were Text as Instruction

- Fluxus artists in the 1960s pioneered the "event score”
  - Short textual instructions that could be performed, interpreted, or simply imagined

- Yoko Ono's "Grapefruit" (1964) collected instruction pieces

CLOCK PIECE

Select a clock.
Set it on time.
You may rewind the clock but never
reset it.
Call it your life clock.
Live accordingly.

1964 spring

THROWING PIECE

Throw a stone into the sky high enough
so it will not come back.

1964 spring

CANNON PIECE

Paste your name on the window.
Borrow a cannon.
Go to a distance and fire against
your name.

The name can be a name or a number
taken out at random from a telephone
book.

If a cannon is not available, you may
use machine-guns, arrows, stones,
spitting, urinating, hose-water, or
any other method.

If nothing is available, go and watch
until the name becomes unrecognizable
by sunset.

You may use telescopes to watch.

1963 autumn

KITE PIECE I

Borrow the Mona Lisa from the gallery.
Make a kite out of it and fly.
Fly it high enough so the Mona Lisa
smile disappears. (a)
Fly it high enough so the Mona Lisa
face disappears. (b)
Fly it high enough so it becomes a dot. (c)

KITE PIECE II

Every year on a certain day, collect
old paintings such as De Kooning, Klein,
Pollack.
Make kites with them and fly.
Fly them high enough and cut the strings
so they will float.

KITE PIECE III

Enlarge your photograph(s) and make many
kites and fly.
When the sky is filled with them, ask
people to shoot.
You may make balloons instead of kites.

1963 autumn

CONCERT PIECE

When the curtain rises, go hide
and wait until everybody leaves
you.
Come out and play.

1963 autumn

HIDE-AND-SEEK PIECE

Hide until everybody goes home.
Hide until everybody forgets about you.
Hide until everybody dies.

1964 spring

Fluxus Event Scores were Text as Instruction

- The score separates the concept from the execution — anyone can perform it, and every performance is different
- This is the direct ancestor of the "prompt"a short text that generates unpredictable creative output

Alison Knowles — The House of Dust (1967)

A HOUSE OF DUST

ON OPEN GROUND

USING NATURAL LIGHT

INHABITED BY FRIENDS AND ENEMIES



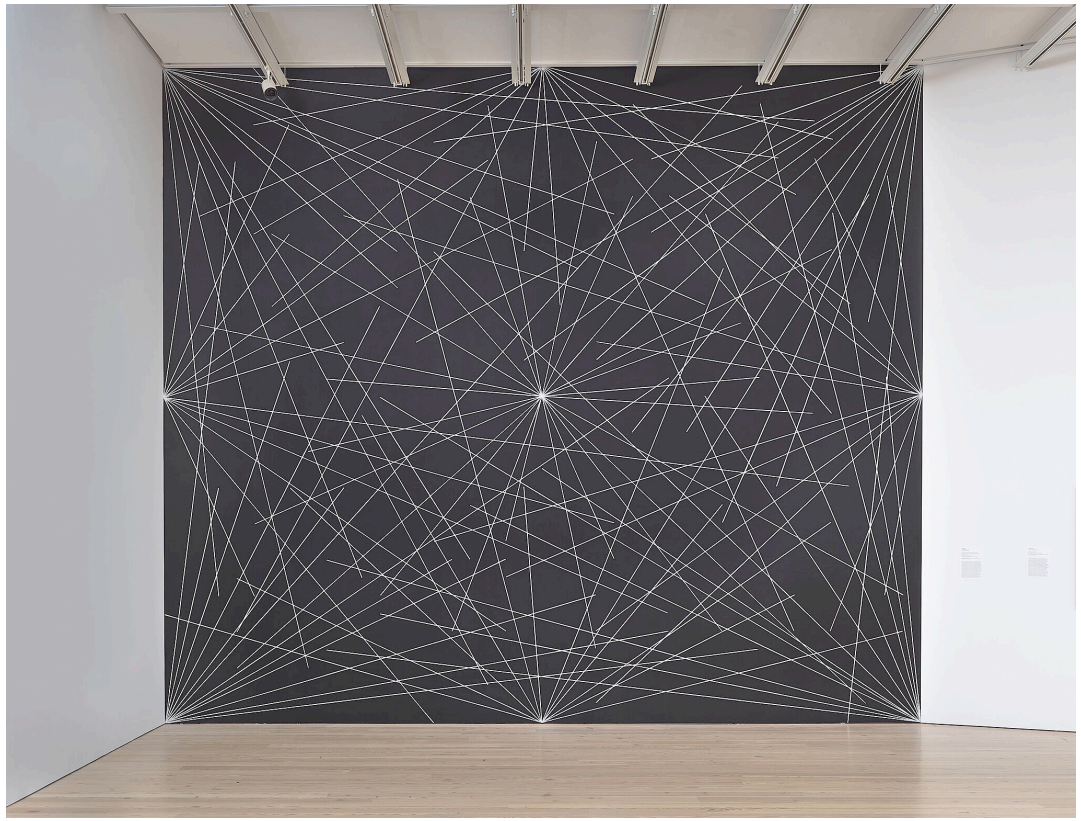
Alison Knowles — The House of Dust (1967)

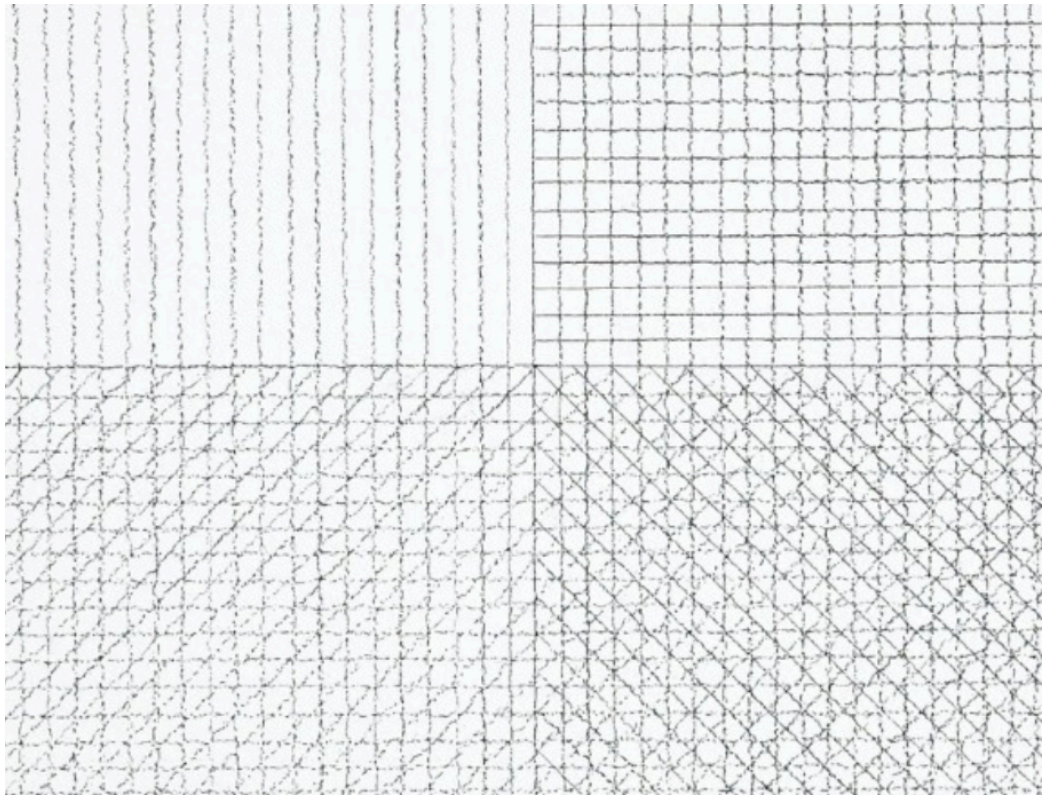
- Fluxus artist Alison Knowles collaborated with composer James Tenney
- Written in FORTRAN on a Siemens mainframe, the program randomly combined four lists: materials, locations, light sources, and inhabitants
  - "A house of [material] / [location] / lit by [light] / inhabited by [inhabitants]"
- From roughly 10,000 possible stanzas, Knowles selected one and translated it into a physical structure built first in Chelsea, New York, and later at CalArts
- The poem becomes instruction, the instruction becomes architecture -> Text generates a chain of creative translations across media

Sol LeWitt — Wall Drawings Instructions as Art (1968–2007)

- Sol LeWitt created over 1,350 wall drawings, but never drew them himself. He wrote text instructions and others executed them
- e.g. Wall Drawing 56 "A square is divided horizontally and vertically into four equal parts, each with lines in four directions superimposed progressively."
- LeWitt compared his instructions to musical scores where each realisation is different depending on who follows them
- There's a direct line from LeWitt's instructions to code-based generative art and to AI prompting

Sol LeWitt — Wall Drawings Instructions as Art (1968–2007)





https://whitney.org/education/families/kids-art-challenge/sol-lewitt

Sol LeWitt Meets ChatGPT Amy Goodchild (2025)

- Artist Amy Goodchild fed Sol LeWitt's wall drawing instructions directly into ChatGPT and asked it to generate p5.js code to execute them
- The results reveal how AI interprets ambiguous human instructions differently from a human draftsperson
- ChatGPT sometimes makes bold, unexpected choices
  - Drawing lines to random locations instead of connecting points
  - Interpreting "irregular" as referring to line length rather than shape
- Highlights the gap between what we mean and what we say as the same ambiguity that made LeWitt's instructions generative in the first place

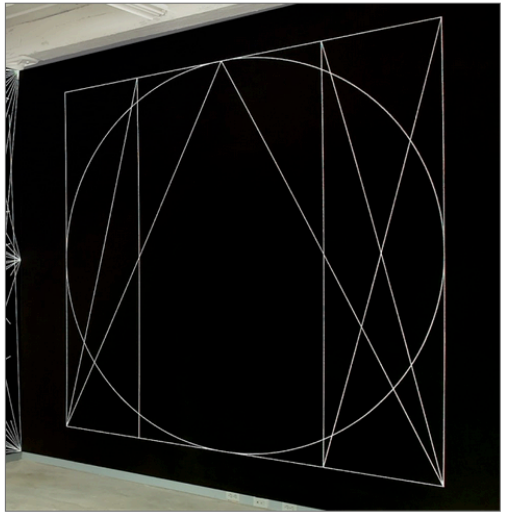
https://www.amygoodchild.com/blog/chatgpt-sol-lewitt-wall-drawings

Sol LeWitt Meets ChatGPT Amy Goodchild (2025)

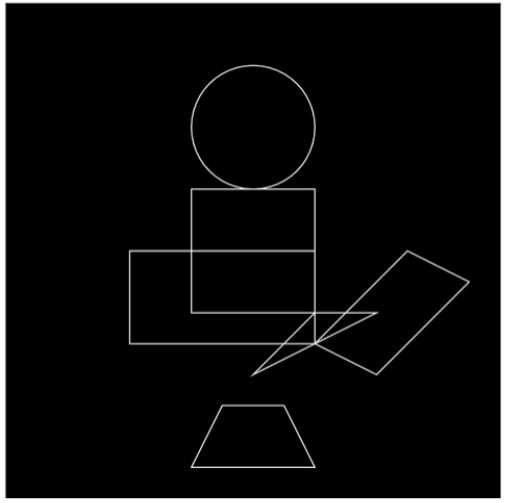
WALL DRAWING #295

"Six white geometric figures (outlines) superimposed on a black wall."

In LeWitt's vocabulary, the six kinds of geometric shapes are: circle, square, triangle, rectangle, trapezoid and parallelogram.



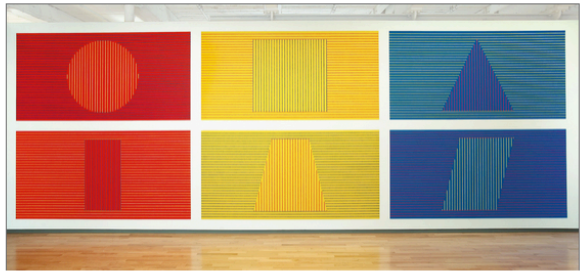
Wall Drawing #295 - Human drawn



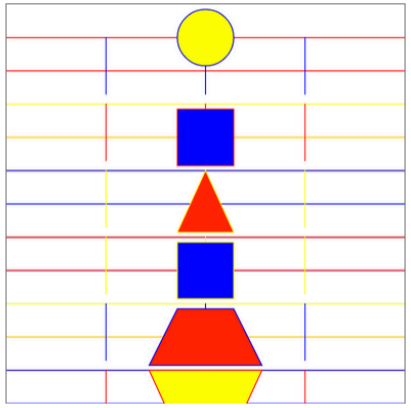
Wall Drawing #295 - ChatGPT output

WALL DRAWING #340

"Six-part drawing. The wall is divided horizontally and vertically into six equal parts. 1st part: On red, blue horizontal parallel lines, and in the center, a circle within which are yellow vertical parallel lines; 2nd part: On yellow, red horizontal parallel lines, and in the center, a square within which are blue vertical parallel lines; 3rd part: On blue, yellow horizontal parallel lines, and in the center, a triangle within which are red vertical parallel lines; 4th part: On red, yellow horizontal parallel lines, and in the center, a rectangle within which are blue vertical parallel lines; 5th part: On yellow, blue horizontal parallel lines, and in the center, a trapezoid within which are red vertical parallel lines; 6th part: On blue, red horizontal parallel lines, and in the center, a parallelogram within which are yellow vertical parallel lines. The horizontal lines do not enter the figures."



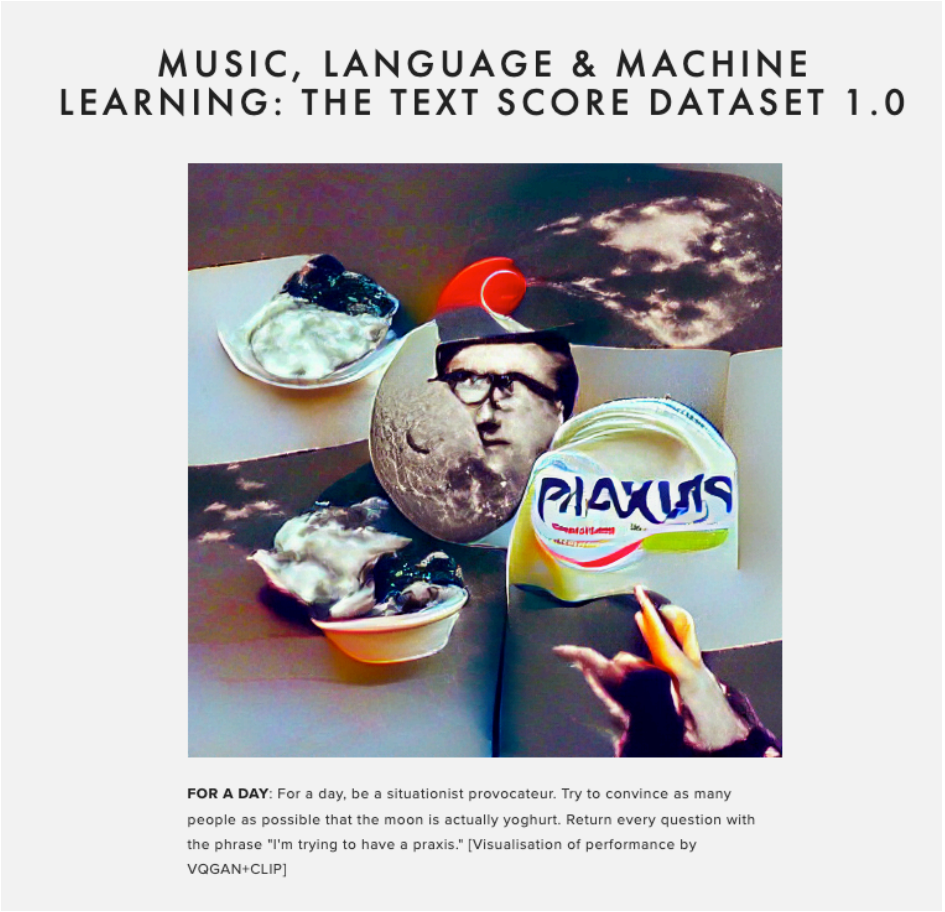
Wall Drawing #340 - Human drawn



Wall Drawing #340 - ChatGPT output

Jennifer Walshe — The Text Score Dataset 1.0 AI-Generated Performance Scores (2021)

- Composer Jennifer Walshe created a corpus of over 3,500 text scores, ranging from Fluxus event scores to works by current practitioners, all painstakingly transcribed by Ragnar Árni Ólafsson
- The dataset was then used to train machine learning algorithms to generate entirely new text scores



Eryk Salvaggio — Fluxus Ex Machina 100 AI Event Scores (2019)

- Artist Eryk Salvaggio fed the Fluxus Performance Workbook (a collection of scores by Yoko Ono, George Brecht, and others) into GPT-2 and generated 100 new event scores
- He curated the outputs, selecting 100 from weeks of generation
- The GPT-2 also independently generated time-interval pieces reminiscent of Cage's 4'33 despite never being trained on Cage's work -> essentially reverse-engineering Fluxus from its descendants!

Eryk Salvaggio — Fluxus Ex Machina 100 AI Event Scores (2019)

- GPT-2 had a "weird propensity toward violence" and produced one score describing a chemical bomb
- Salvaggio later fed his AI-generated scores into DALL-E 2 as prompts, creating fictional documentary photographs of the performances
- He notes that the project became "almost impossible to create using GPT-3" because the newer model was too good at eliminating ambiguity

Text to Everything

- Generated text now routinely feeds into other generative systems: text-to-image, text-to-music (MusicLM, Udio), text-to-video (Sora), text-to-3D
- This changes the role of generated text from final output to intermediate instruction as language becomes a control interface for all media